f) $$\alpha_{absorbed} = \frac{24 V \ln 10}{c\, S_{absorbed}} \left(\frac{1}{T_{60,med}} - \frac{1}{T_{60,mean}} \right)$$

a) V: hier ist 60x unabhängiger miteinander

$$\sigma_\alpha^2 = \left(\frac{\partial \alpha}{\partial T_{med}} \sigma_{T_{med}} \right)^2 + \left(\frac{\partial \alpha}{\partial T_{mean}} \sigma_{T_{mean}} \right)^2$$

$$= \left(- \frac{k}{T_{med}^2} \sigma_{T_{med}} \right)^2 + \left(\frac{k}{T_{mean}^2} \sigma_{T_{mean}} \right)^2$$

$$= k \sqrt{\frac{\sigma_{med}^2}{T_{med}^4} + \frac{\sigma_{mean}^2}{T_{mean}^4}}$$

b) $$\sigma_\alpha = \frac{24 V \ln 10}{c\, S_{absorbed}} \sqrt{\frac{\sigma_{T_{med}}^2}{T_{med}^4} + \frac{\sigma_{T_{mean}}^2}{T_{mean}^4}}$$

$$= \frac{24 \cdot 240 \cdot \ln 10}{343{,}4 \cdot 10}$$

Mittelwerte:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$\bar{T}_{med} = \frac{1}{8} \sum \text{Datensatz} = 3{,}526$$

$$\bar{T}_{mean} = \frac{1}{8} \sum \text{Datensatz} = 4{,}154$$

Standarddaten

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$$\sigma_{T_{med}} = \sqrt{\frac{1}{7} \sum (x_{i0} - \bar{x})^2} = 0.3244$$

$$\sigma_{T_{mean}} = \sqrt{\frac{1}{7} \sum (x_{i0} - \bar{x})^2} = 0.1792$$

$$\sigma_\alpha = \frac{24 \cdot 240 \cdot \ln 10}{343{,}4 \cdot 10} \sqrt{\frac{(0.3244)^2}{(3.526)^4} + \frac{(0.1792)^2}{(0.1792)^4}}$$

$$= 0.108$$

c) $$\Rightarrow \bar{x} \pm t_{0.975, n-1} \cdot \frac{\sigma}{\sqrt{n}}$$

→ oder 3 wie σ₀ ist

$$\bar{x} \pm 2 \frac{\sigma}{\sqrt{n}}$$

$$\bar{x} = 0.165$$

Nebr.:

→ tabellierte 10.02,96

$$0.165 - 2{,}36 \frac{0.108}{\sqrt{8}}$$

$$0.165 - 2{,}36 \frac{0.108}{\sqrt{8}}$$

$$\Rightarrow [0.075, 0.256]$$

2) $y = kt$

minimiser ei

$$\varepsilon = \sum (y - y_i)^2$$

$$= \sum (kt_i - y_i)^2$$

$$\frac{d\varepsilon}{dk} = \sum 2(kt_i - y_i)\, t_i$$

$$= 2k \sum t_i - 2 \sum y_i t_i$$

$$= 0$$

$$k = \frac{\sum t_i y_i}{\sum t_i^2}$$

3) $x = x_{sann} + \xi \qquad \sim N(0, \sigma_x)$

$y = y_{sann} + \eta \qquad \sim N(0, \sigma_y)$

a) $d = (y_{sann} + \eta) - (x_{sann} + \xi)$

$d = y - x$

$y_{sann} - x_{sann}$

$= d_{sann} + \eta - \xi$

↳ sil : $d = \eta - \xi$

$\mu_\eta = \mu_\xi = 0$

$\mu_d = \mu_{sann}$

ja, forvented verdi er lik den sanne distansen

b) sisen $d = -x$

$$\sigma_d^2 = \left(\frac{\partial d}{\partial y}\right)^2 \sigma_y^2 + \left(\frac{\partial d}{\partial x}\right)^2 \sigma_x^2$$

$$= 1 \cdot \sigma_y^2 + \sigma_x^2 \cdot 1$$

$$= \sigma_y^2 + \sigma_x^2$$

$$= 2\sigma_x^2$$

$$Var(d) = Var(\eta - \xi)$$

$$= Var(\eta) - Var(\xi)$$

$$= \sigma_x^2 + \sigma_x^2$$

$$= 2\sigma_x^2$$

$$Std = \sqrt{2}\, \sigma_x$$

c) $d = \sqrt{(y-x)^2}$

$d = |y - x|$

$\frac{\partial d}{\partial y} = \frac{y-x}{\sqrt{(y-x)^2}} = 1$ } bruker ikke

$\sigma_d = \sqrt{2}\,\sigma_x$

d) $d^2 = (y-x)^2$

$= (y_{true} + n - x_{true} - \xi)^2$

$= (y_{true} - x_{true} + \xi)^2$ $\quad \xi = n - \xi$, $\xi \sim N(0, \sqrt{2}\sigma_x)$

$= (y_{true} - x_{true})^2 + 2(y_{true} - x_{true})\xi + \xi^2$

Forventningsrett

$E[d^2] = d^2_{true} + 2d_{true} \underbrace{E[\xi]}_{=0} + \underbrace{E[\xi^2]}_{Var(\xi)}$

$= d^2_{true} + Var(\xi)$

$= d^2_{true} + 2\sigma_x^2$

$\hat{d} = \sqrt{d^2}$

$= \sqrt{d^2_{true} + 2\sigma_x^2}$

$\approx \sqrt{d^2_{true} + 2\sigma_x^2}$

Denne modellen er ikke forventningsrett